Introduction of Financial Professional Certificate



Homework_Financial Reporting and Analysis
Q3 answers

Formulas

- $\frac{\text{current assets}}{\text{current liabilities}}$
- $\frac{\text{total debt}}{\text{total equity}}$
- $\frac{\text{EBIT}}{\text{interest payments}}$
- $\frac{\text{gross profit}}{\text{revenue}}$
- $\frac{\text{net income} - \text{preferred dividends}}{\text{average common equity}}$
- $\frac{\text{EBIT}}{\text{short and long term debt + equity}}$
- $\frac{\text{net income}}{\text{revenue}}$
- $\frac{\text{cash+ recivables+ marketable securiti}}{\text{current liabilities}}$
- $\frac{365}{\text{revenue / average receivables}}$
- $\frac{\text{cost of goods sold}}{\text{average inventories}}$
- $\frac{\text{revenue}}{\text{average assets}}$
- $\frac{\text{revenue}}{\text{average working capital}}$

Current Ratio

- $\text{current ratio} = \frac{\text{current assets}}{\text{current liabilities}}$

- $\text{current ratio} = \frac{620}{325} = 1.9$

- The current ratio indicates lower liquidity levels when compared to last year and more liquidity than the industry average.

Quick Ratio

- $\text{quick ratio} = \frac{\text{cash+ recivables+ marketable securities}}{\text{current liabilities}}$

- $\text{quick ratio} = \frac{(105+205)}{325} = 0.95$

- The quick ratio is lower than last year and is in line with the industry average.

Days of Sales Outstanding

- DSO (days of sales outstanding) $= \dfrac{365}{\text{revenue / average receivables}}$

- $\text{DSO} = \dfrac{365}{4{,}000/\,[(205+195)/2]} = 18.25$

- The DSO is a bit lower relative to the company's past performance but slightly higher than the industry average.

Inventory Turnover

- $\text{inventory turnover} = \dfrac{\text{cost of goods sold}}{\text{average inventories}}$

- $\text{inventory turnover} = \dfrac{3{,}000}{(310+290)/2} = 10.0$

- Inventory turnover is much lower than last year and the industry average. This suggests that the company is not managing inventory efficiently and may have obsolete stock.

Total Asset Turnover

- $\text{total asset turnover} = \dfrac{\text{revenue}}{\text{average assets}}$

- $\text{total asset turnover} = \dfrac{4{,}000}{(2.060 + 1.940)/2} = 2.0$

- Total asset turnover is slightly lower than last year and the industry average.

Working Capital Turnover

- working capital turnover $= \frac{\text{revenue}}{\text{average working capital}}$

- beginning working capital $= 580 - 275 = 305$
- ending working capital $= 620 - 325 = 295$

- working capital turnover $= \frac{4{,}000}{(305 + 295)/2} = 13.3$

- Working capital turnover is lower than last year, but still above the industry average.

Gross Profit Margin

- $\text{gross profit margin} = \frac{\text{gross profit}}{\text{revenue}}$
- $\text{gross profit margin} = \frac{1{,}000}{4{,}000} = 25\%$
- The gross profit margin is lower than last year and much lower than the industry average.

Net Profit Margin

- $\text{net profit margin} = \frac{\text{net income}}{\text{revenue}}$
- $\text{net profit margin} = \frac{200}{4{,}000} = 5.0\%$
- The net profit margin is lower than last year and much lower than the industry average.

Return on Total Capital

- return on total capital $= \dfrac{\text{EBIT}}{\text{short and long term debt + equity}}$

- beginning total capital $= 140 + 45 + 690 + 880 = 1.755$
- ending total capital $= 160 + 55 + 610 + 1{,}020 = 1.845$

- return on total capital $= \dfrac{350}{(1{,}755+1{,}845)/2} = 19.4\%$

- The return on total capital is below last year and below the industry average. This suggests a problem stemming from the low asset turnover and low profit margin.

Return on Common Equity

- return on common Equity $= \frac{\text{net income} - \text{preferred dividends}}{\text{average common equity}}$

- return on common Equity $= \frac{200}{(1{,}020+880)/2} = 21.1\%$

- The return on equity is lower than last year but better than the industry average. The reason it is higher than the industry average is probably because greater use of leverage.

Debt-to-Equity Ratio

- $\text{debt} - \text{to} - \text{equity ratio} = \frac{\text{total debt}}{\text{total equity}}$

- $\text{debt} - \text{to} - \text{equity ratio} = \frac{610+160+55}{1{,}020} = 80.9\%$

- The debt to equity ratio is lower than last year but still much higher than the industry average. This suggests the company is trying to get its debt level more in line with the industry.

Interest Coverage

- $\text{interest coverage} = \dfrac{\text{EBIT}}{\text{interest payments}}$

- $\text{interest coverage} = \dfrac{350}{50} = 7.0$

- The interest coverage is better than last year but still worse than the industry average. This, along with the slip in profit margin and return on assets, might cause some concern.